

A4 Understanding Linear Discriminant Analysis - Answers

Understanding Linear Discriminant Analysis (LDA)

1. LDA may perform poorly when:

- ☐ A) Classes are well separated
- ☐ B) Data satisfies Gaussian assumptions
- ☐ C) Classes have significant nonlinear boundaries
- ☐ D) Features are informative

Answer: C) LDA is linear and may perform poorly when class boundaries are strongly nonlinear.

2. Which assumption of LDA is often violated in real -world datasets?

- ☐ A) Equal covariance matrices across classes
- ☐ B) Presence of labels
- ☐ C) Numerical features
- ☐ D) Finite samples

Answer: A) Equal covariance matrices across classes is often violated.

3. LDA can struggle when:

- ☐ A) The number of samples is much larger than the number of features
- ☐ B) The number of features exceeds the number of samples
- ☐ C) There are only two classes
- ☐ D) Features are standardized

Answer: B) When features outnumber samples, covariance/scatter estimates may be singular or unstable.

4. Why can LDA be sensitive to outliers?

- ☐ A) It uses class means and covariance estimates
- ☐ B) It performs clustering
- ☐ C) It ignores feature values
- ☐ D) It uses decision trees

Answer: A) Class means and covariance estimates are sensitive to extreme observations.

5. Which of the following is a limitation of LDA?

- ☐ A) Requires labeled data
- ☐ B) Assumes normally distributed classes
- ☐ C) Assumes equal class covariances
- ☐ D) All of the above

Answer: D) Classical LDA requires labels and commonly assumes Gaussian classes with shared covariance.

6. When class covariance matrices are substantially different, which method may outperform LDA?

- ☐ A) Quadratic Discriminant Analysis (QDA)
- ☐ B) PCA

- o C) K -Means
- o D) KNN Regression

Answer: A) QDA allows class-specific covariance matrices and may fit such data better.

7. LDA assumes that class boundaries are:

- o A) Linear
- o B) Circular
- o C) Arbitrary
- o D) Fractal

Answer: A) Classical LDA produces linear decision boundaries.

8. What problem arises when the within -class scatter matrix S_W is singular?

- o A) Projection becomes easier
- o B) Matrix inversion becomes difficult or impossible
- o C) Classes become balanced
- o D) Dimensionality increases automatically

Answer: B) A singular S_W is difficult or impossible to invert directly.

9. LDA may not effectively separate classes when:

- o A) The classes are linearly separable
- o B) The classes overlap heavily
- o C) Features are relevant
- o D) Labels are correct

Answer: B) Heavy overlap limits the discriminative information available to a linear projection.

10. In high -dimensional datasets with few samples, LDA often suffers from:

- o A) Overfitting and unstable covariance estimates
- o B) Underfitting only
- o C) Reduced computation time
- o D) Automatic feature selection

Answer: A) High dimensionality with few samples can cause overfitting and unstable covariance estimates.

11. Which data characteristic can negatively impact LDA performance?

- o A) Strong outliers
- o B) Noisy measurements
- o C) Non -Gaussian distributions
- o D) All of the above

Answer: D) Outliers, noise, and non-Gaussian distributions can all reduce performance.

12. LDA is generally less suitable than kernel -based methods when:

- o A) Data is linearly separable
- o B) Data contains nonlinear class structures
- o C) Data has few features

- o D) There are only two classes

Answer: B) Kernel methods can represent nonlinear class structure.

13. A major limitation of LDA compared to PCA is that:

- o A) It requires class labels
- o B) It can handle more dimensions
- o C) It is unsupervised
- o D) It ignores covariance

Answer: A) Unlike PCA, LDA needs class labels.

14. Why might class imbalance affect LDA?

- o A) Larger classes can dominate mean and covariance estimation
- o B) LDA automatically balances classes
- o C) LDA ignores sample counts
- o D) Class imbalance improves separability

Answer: A) Larger classes can dominate class statistics unless the procedure accounts for imbalance.

15. For a dataset with C classes, a limitation of LDA is that the maximum number of discriminant components is:

- o A) Equal to the number of samples
- o B) Equal to the number of features
- o C) $C-1$
- o D) Unlimited

Answer: C) The between-class scatter rank limits components to $C - 1$.

16. Which statement best describes a limitation of classical LDA?

- o A) It assumes a common covariance structure across all classes
- o B) It works perfectly for any nonlinear dataset
- o C) It does not require labels
- o D) It is immune to outliers

Answer: A) Classical LDA assumes a common covariance structure across classes.

17. LDA may produce suboptimal results when:

- o A) Feature distributions deviate significantly from normality
- o B) Classes are clearly separated
- o C) Data is low -dimensional
- o D) Covariances are equal

Answer: A) Strong departures from Gaussian class distributions can make the model suboptimal.

18. Which of the following is known as the "small sample size problem" in LDA?

- o A) Too many classes compared to features
- o B) Too many features compared to samples
- o C) Too many samples compared to features

o D) Too few labels

Answer: B) The small-sample-size problem occurs when features greatly outnumber samples.

19. LDA cannot extract more than $C-1$ discriminant features because:

o A) The rank of the between -class scatter matrix is limited

o B) NumPy restricts dimensions

o C) Eigenvalues are always zero

o D) PCA removes extra dimensions

Answer: A) Between-class scatter has rank at most $C - 1$.

20. Which scenario is most likely to violate LDA assumptions?

o A) Two Gaussian classes with equal covariance matrices

o B) Multiple classes with highly nonlinear boundaries and unequal covariances

o C) Balanced classes with low noise

o D) Well -separated classes in low dimensions

Answer: B) Nonlinear boundaries and unequal covariances violate key classical LDA assumptions.